

In the ever-evolving world of software development, data lies at the heart of every application. Whether you're building a high-traffic e-commerce platform, a lightweight web app, or an enterprise-level analytics engine, your choice of database can make or break your project. Among the multitude of options, three names dominate the open source relational database landscape: PostgreSQL, MySQL, and MariaDB. Each of these RDBMS (relational database management systems) has carved out a loyal user base and distinct identity, offering a mix of speed, reliability, extensibility, and community support.

The story of open source databases begins with the need for freedom — freedom from vendor lock-in, freedom to customise, and freedom to innovate. While commercial databases like Oracle and SQL Server once ruled the enterprise world, developers around the globe began to shift towards more flexible and cost-effective alternatives.

MySQL, created in 1995, quickly became a favourite due to its simplicity, performance, and ease of integration with the LAMP stack. PostgreSQL, although older (dating back to the 1980s as the 'POSTGRES' project), steadily gained traction for its advanced features and standards compliance.

Enter MariaDB — born out of concern when Oracle acquired Sun Microsystems (and with it, MySQL) in 2010. Michael 'Monty' Widenius, one of MySQL's original creators, forked the project to ensure that the community retained full control and transparency.

This sparked a new era of competition and innovation in the open source database world, giving rise to three strong contenders — each with its own philosophy, community, and technical direction.

- **PostgreSQL** is often dubbed the "world's most advanced open source database." Known for its powerful support for complex

queries, full ACID compliance, and standards-based architecture, PostgreSQL is the go-to choice for applications where data integrity and extensibility are paramount

- **MySQL**, now owned by Oracle, remains one of the most widely adopted databases globally. It is synonymous with speed, simplicity, and ease of use. Many popular platforms, including WordPress and Drupal, rely on MySQL under the hood.
- **MariaDB** positions itself as a drop-in replacement for MySQL but has evolved into a distinct database with its own performance enhancements, storage engines, and security features. Driven by a passionate open source community, MariaDB offers both compatibility and innovation.

Despite their shared roots in relational database principles, these three systems differ significantly in design philosophy, feature sets, licensing, and use cases — making the choice between them far from trivial.

The contenders: A quick profile

MySQL: The web application workhorse

If you've ever built or run a website, chances are you've interacted with MySQL — either directly or through a content management system like WordPress, Joomla, or Drupal. MySQL earned its place in the spotlight by offering a fast, reliable, and easy-to-use solution for web applications. It became a cornerstone of the famous LAMP stack (Linux, Apache, MySQL, PHP), making it a default choice for countless developers and hosting providers.

MySQL's strengths lie in its speed, simplicity, and mature ecosystem. It handles read-heavy workloads like blog posts, product catalogues, and dashboards with ease. Over the years, Oracle's stewardship has added

enterprise features, improved security, and maintained compatibility — but it's also led some developers to question its open source commitment. Nevertheless, with massive global adoption and a robust support infrastructure, MySQL remains a go-to database for small to medium-sized web applications that prioritise speed and ease of deployment.

MariaDB: The community- driven alternative

Born from the same DNA as MySQL, MariaDB is more than just a clone — it's a community-led evolution. Created by the original developers of MySQL in response to Oracle's acquisition, MariaDB promised to stay truly open, transparent, and innovative. And it was delivered. While maintaining drop-in compatibility with MySQL (especially in earlier versions), MariaDB has introduced its own set of optimisations, unique features, and performance-focused storage engines like Aria and ColumnStore.

MariaDB is known for its commitment to open source, its frequent release cycle, and its cutting-edge improvements in speed, scalability, and replication. Many Linux distributions, including Fedora and Arch, have even replaced MySQL with MariaDB as their default RDBMS. For developers and organisations who value openness and community-driven development — without sacrificing performance — MariaDB is an attractive and forward-thinking choice.

PostgreSQL: The feature-rich powerhouse

If MySQL and MariaDB are built for speed and simplicity, PostgreSQL is designed for power, precision, and possibility. Often referred to as the most advanced open source relational database in the world, PostgreSQL (or 'Postgres') is a favourite among developers who demand full SQL compliance, support for complex queries, and extensibility.